Hard Margin SVM – Mathematics

✓ We already know from the **geometric intuition**:

✓ SVM tries to find the **best hyperplane** that separates classes with the **maximum margin**.

✓ Now let's put that into **mathematical form**.

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1 Equation of a Hyperplane

For n-dimensional input **x**, the hyperplane can be written as:

$$w^T x + b = 0$$

- **w** = weight vector (normal to the hyperplane)
- **b** = bias (shifts the hyperplane)
- **x** = input vector

The sign of $w^T x + b$ determines the class:

- > 0 → Class +1
- < 0 → Class -1

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2 Margin Definition

- Margin = distance between the hyperplane and the **closest data points (support vectors)**.
- Distance of a point (x_i, y_i) from the hyperplane:

$$\text{distance} = \frac{|w^T x_i + b|}{||w||}$$

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3 Optimization Goal

We want to **maximize the margin**.

- Margin is **2 / ||w||** (distance between support vectors of +1 and -1).

So maximizing margin ↔ minimizing $||w||$.

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4 Constraints (for correct classification)

For each training example (x_i, y_i):

- $y_i(w^T x_i + b) \geq 1$

Where $y_i \in \{+1, -1\}$.

This ensures:

- Positive points lie on one side
- Negative points lie on the other side

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5 Final Optimization Problem

The hard margin SVM becomes:

$$\min_{w,b} \frac{1}{2} \|w\|^2$$

subject to:

$$y_i(w^T x_i + b) \geq 1 \quad \forall i$$

- Objective: minimize $\|w\|^2$ (maximize margin).
- Constraint: correctly classify all training points.

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6 Why is this Quadratic Optimization?

- The objective function $\frac{1}{2}\|w\|^2$ is **quadratic**.
- The constraints $y_i(w^T x_i + b) \geq 1$ are **linear**.

  👉 Hence, it's a **Quadratic Programming (QP)** problem.

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7 Support Vectors in Math

- Only points that **exactly satisfy** $y_i(w^T x_i + b) = 1$ are **support vectors**.
- These points directly define the optimal hyperplane.

Quick Recap

- Equation of hyperplane: $w^T x + b = 0$
- Margin = $\frac{2}{||w||}$
- Optimization: minimize $\frac{1}{2}\|w\|^2$
- Constraint: $y_i(w^T x_i + b) \geq 1$
- Support vectors are the closest points that **touch the margin boundary**.
